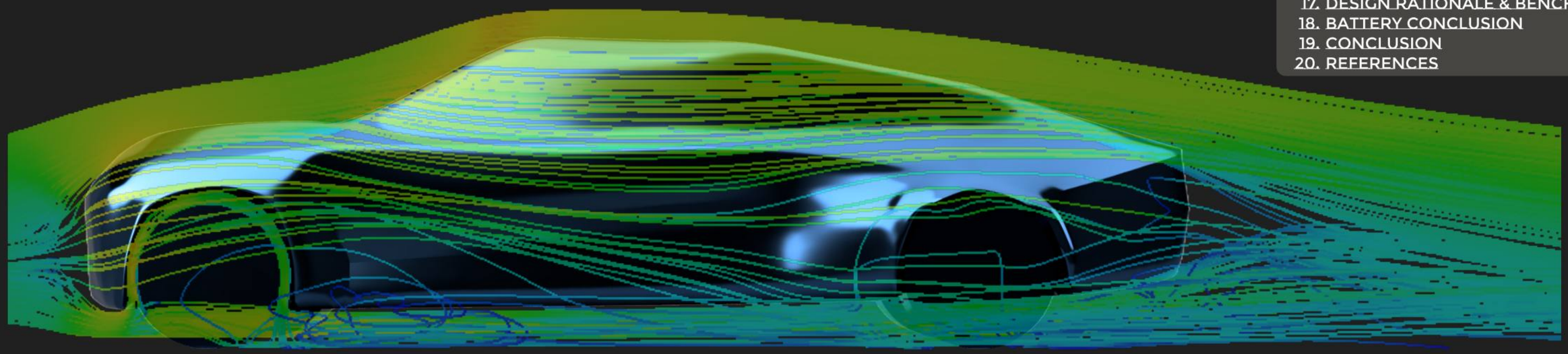
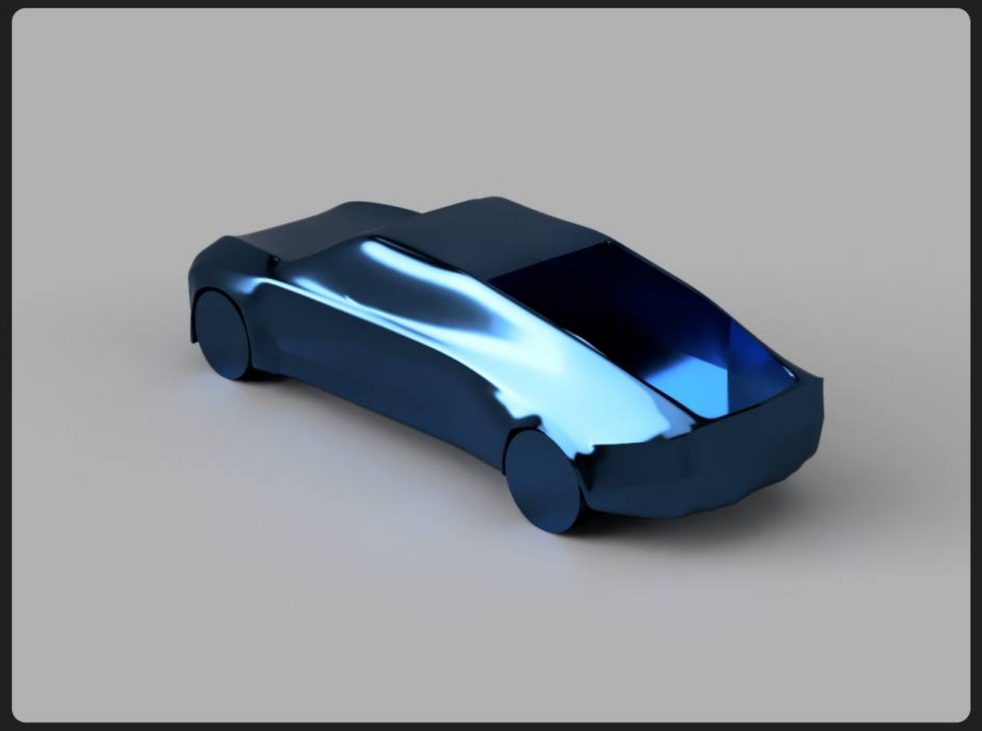
VENTARA
TRUCK

VEHICLE
1. COVER
2. DESIGN SUMMARY
3. DESIGN RATIONALE
4. BENCHMARKING
5. SKETCHING
6. MODEL ITERATING
7. CAD
8. FRONTAL AREA MEASUREMENT
9. CFD SETUP
10. MESHING
11. 2D PRESSURE & VELOCITY
12. 3D PRESSURE & VELOCITY
13. AERODYNAMIC FORCES & COEFFICIENTS
14. ANALYSIS OF RESULTS
15. FUTURE PROPOSALS FOR IMPROVEMENT
16. PIPELINE RESEARCH & ITERATING
17. DESIGN RATIONALE & BENCHMARKING
18. BATTERY CONCLUSION
19. CONCLUSION
20. REFERENCES



Yousuf
Shahabuddin

DESIGN SUMMARY



Ventara is a pick-up truck designed for long highway drives that don't break the bank, whilst holding everything you could need in its bed. The aerodynamics of the car do successfully demonstrate its efficiency, which is comparable to sports SUVs and high-end pick-up trucks.

DRAG COEFFCIENT: 0.3864

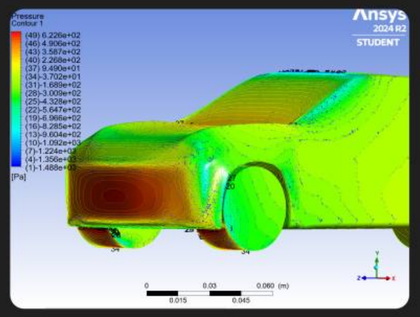
DRAG FORCE: 293.87 N

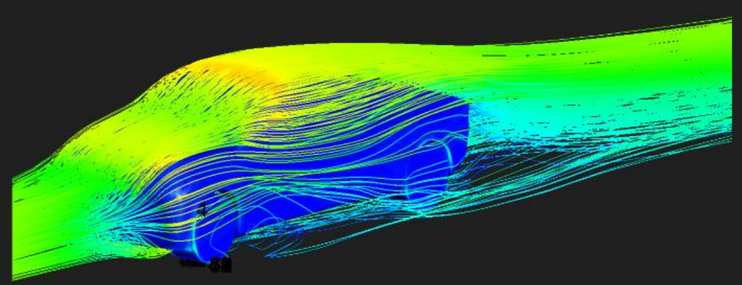
LIFT COEFFICIENT: 0.2396

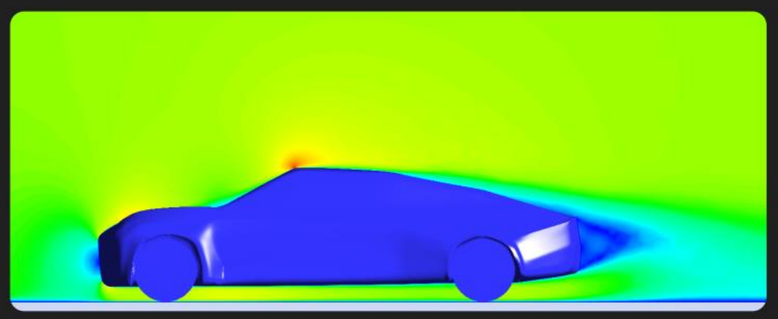
DOWNFORCE: -182.09N

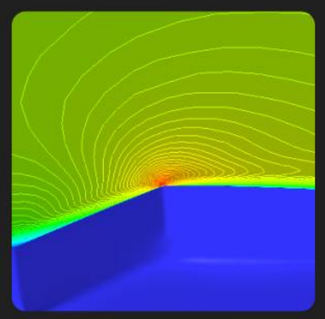
BATTERY COOLING PLATE

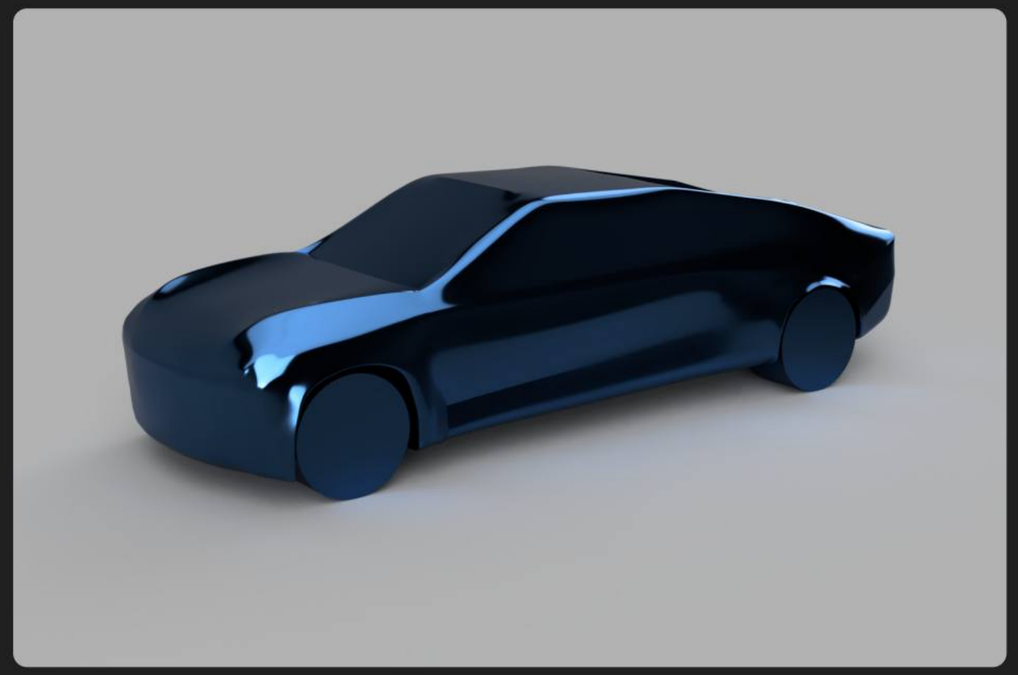










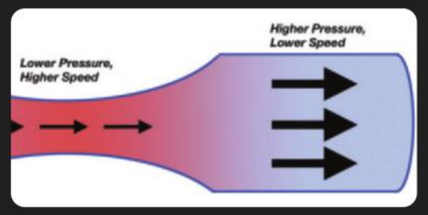


DESIGN RATIONALE

Pickup trucks are the most common car type in the USA and middle east [2]. Heavy payloads, frequent usage, and long distance driving mean that any marginal improvements in efficiency will add up significantly. For example, in aircraft, the 777-X is approximately 15% more efficient than its predecessor which was launched 25 years prior, but the new one has raked in $250bn in revenue [3].

Curved body

Creasing is kept to a minimum. Whilst creases serve a stylistic purpose on the sides of the car, they create high pressure points as well as higher turbulence eddy dissipation. Thus, the car has been rounded where suitable in order to improve the drag effects without drastically hurting its aesthetics.



Sloped bed

Having the tailgate up actually decreases drag compared to having it
lowered on an open bed. It creates a pocket of air that has low
pressure and circulates within the bed, trapping the air. Instead of removing the tailgate or reshaping it vastly, sloped bed walls and back will help guide airflow smoothly over and down the car.


[6]

Teardrop

A teardrop shape is the most aerodynamic design [4] for a car. It smoothens airflow, as the rounded front reduces pressure, while the rear ended tapers off gently, minimising flow
separation. The air will flow around the car, vastly reducing the drag caused by high pressure.



[7]

Rear diffuser

Drawing from Bernoulli's principle, high velocity airflow will result in lower pressure at the rear, generating downforce [5]. A diffuser will expand that area, increasing the area for air to flow and thus decreasing the velocity, recovering pressure for downforce. Low pressure behind a car acts as drag, essentially pulling the car back. So a smaller wake will equal lower pressure drag. Additionally, exhaust gases within this hot air from cooling will decrease the pressure. so merging the exhausts with this has some benefit.

Creasing

Longitudinal creases on the bonnet are both for airflow and style. They guide the air towards the A-pillar, and keep the flow attached as it moves over to the windscreen, reducing flow separation and guiding the air to where we need it to go.

Sculpted sides

Channels air for smooth flow and a cleaner exit. These contours are aesthetically pleasing whilst also serving functionally to reduce turbulence, directing air away from the wheels. The idea is to be the least obtrusive as possible, flowing with the air rather than opposing it.

BENCHMARKING

CYBERTRUCK


[8]

Tesla Cybertruck has a unique sloped roof, that transitions smoothly into the truck-bed, allowing for smoother airflow over the vehicle.

Its sharp edges create high pressure points which do increase the drag coefficient, however this was done for aesthetic purposes and to reduce manufacturing processes. The removable truck bed cover allows for the extra space when needed, and improved efficiency otherwise.

Whilst the Cybertruck is more aerodynamic that some pick-up trucks, its flat panels create huge frontal pressure. Its side panel are not contoured either, and with no redirection of air, high velocity vortices form as a result.

Inspiration from the Rolls Royce drop tail for the rear section- curves inwards and down to guide air over the car smoothly, reducing wakes. Whilst akin to a roadster, this aerodynamic feature can be applied to the bed of pick-up trucks.

DROP TAIL


[11]

The McLaren Speedtail is renowned for its aerodynamics. With a drag coefficient of just 0.278, it is one of the most aerodynamic cars, bar the Mercedes EQS at 0.20, although that has a lower top speed and downforce.
The Speedtail fully exploits the teardrop shape, tapering off into a long tail, with a spoiler built into the body of the car. Aero hub caps have also been specially designed to direct the air over the wheels, through the contoured body panels and out the back. Seating however, is awkward due to the teardrop shape and rounded cockpit - driver sits in the middle, with passengers either side and shifted back.

SPEEDTAIL


[9]


[9]

The Rivian R1T is the most aerodynamic pick-up truck created. As an EV, efficiency is imperative as it needs to compete against trucks that have huge fuel tanks. One of the benefits of being an EV is its flat underbody; less cooling is required as there is no combustion, and battery packs are often flat. This allows air to flow smoothly below the car. Its drag coefficient of 0.30 is aided by front air curtains that create an “air wall” around the front wheels - designed to take high pressure air from the front stagnation region and accelerate it into high velocity jets that bypass the wheel interface.

Although it does improve on efficiency, it still has exposed wheels and operates with the geometric constraints of a pick-up truck. Corners in the roof structure can be rounded, and fewer drivetrain components should be exposed. These vehicles need high ground clearance, but the wheel arches do not need to leave the wheels fully exposed - an enclosure more in line with the rest of the chassis would be preferential. Side mirrors increase drag, and can be replaced be cameras, as the Mclaren Speedtail has done. Further gains appear achievable through enhanced rear pressure recovery and more aggressive wheel wake management.

Smooth body reduces skin friction drag, and lowers base drag at the rear.

RIVIAN R1T


[12]

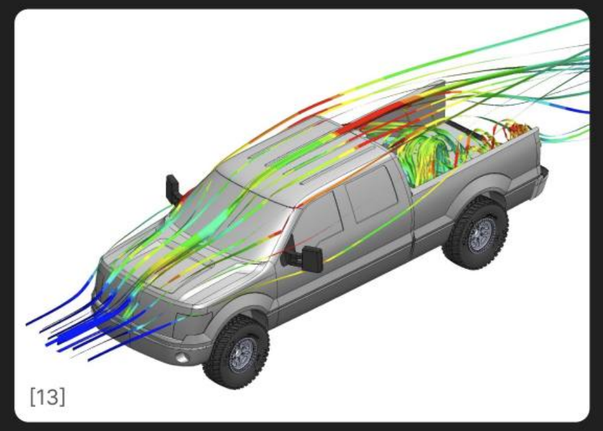
Smooth body reduces skin friction drag, and lowers base drag at the rear.

TOURBILLON

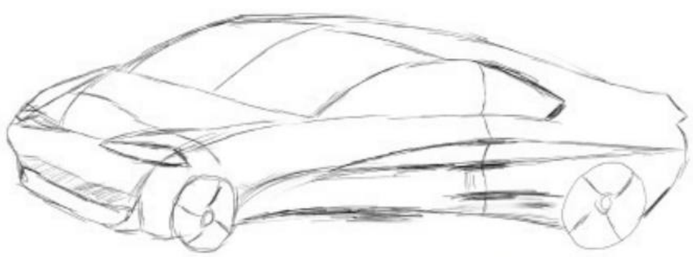

[10]

The Bugatti Tourbillon uses a large underfloor diffuser integrated with a flat floor and venturi system. It acts as a pressure recovery device, allowing accelerated underbody flow to expand gradually, reducing separation and increasing base pressure. This generates strong rear downforce with lower drag penalties than a rear wing alone. In high-speed mode, with reduced ride height and a retracted wing, the diffuser becomes the primary source of aerodynamic stability, highlighting the importance of expansion angle and ride-height integration.

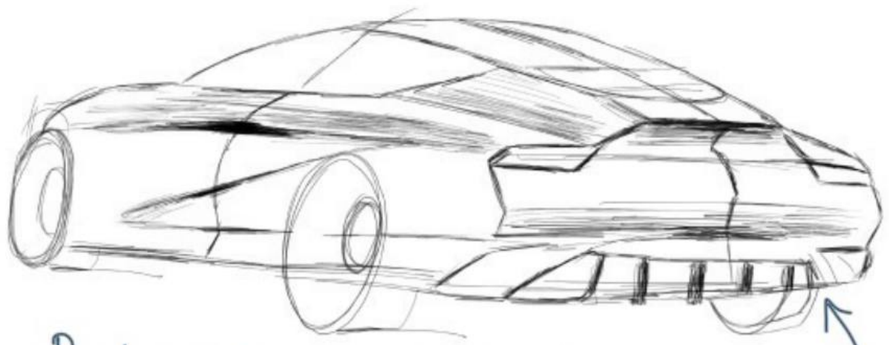
Pick-up trucks, despite having a bed which decreases efficiency, having the tailgate closed is better than having it open. Whilst an open tailgate lengthens the car, it disrupts a vital air cushion created by the enclosed bed. A low pressure bubble is created, wherein air is circulated. Therefore I will keep my truck-bed closed to preserve this, allowing air to flow directly over it for the most part.


[13]

SKETCHING



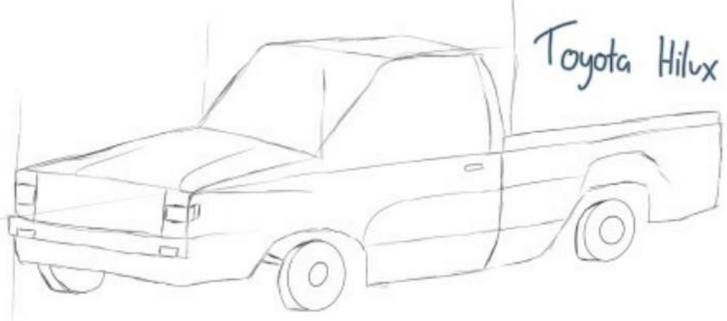
Initial sketches



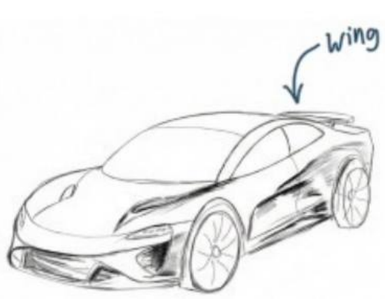
Porsche rough sketch + taillight adaptation



sketch of an ai generated car



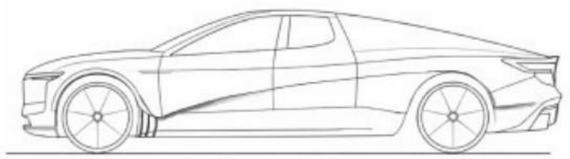
Toyota Hilux



wing for downforce



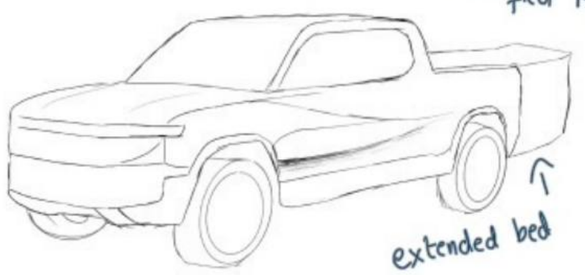
Audi RSQ8 inspiration





orthographic views

Modified Rivian



extended bed

front splitter for downforce

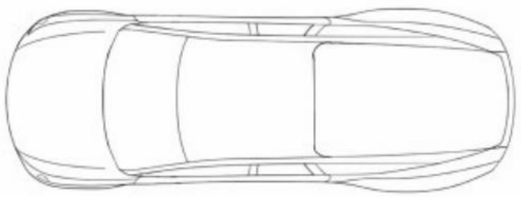


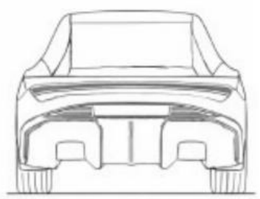
Take on Ford Raptor

smoothens wake

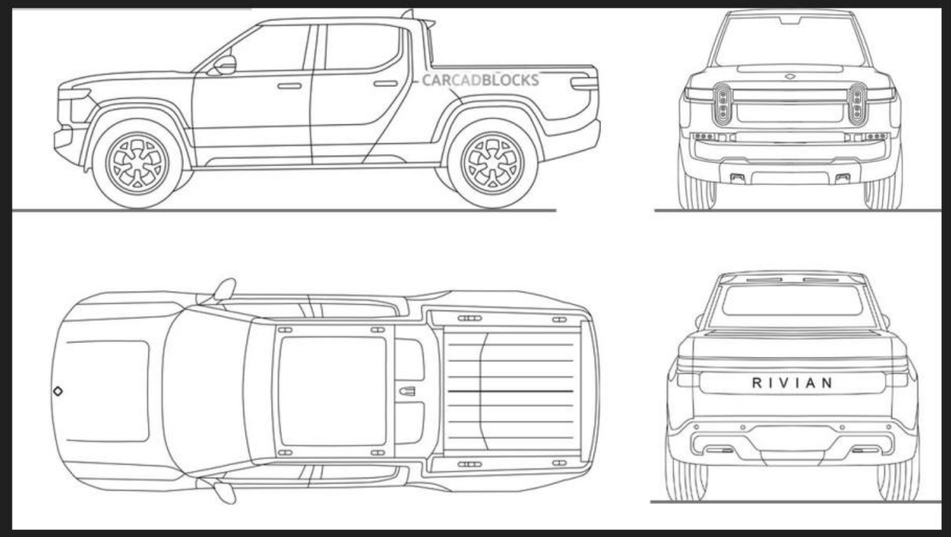


Sports pick-up



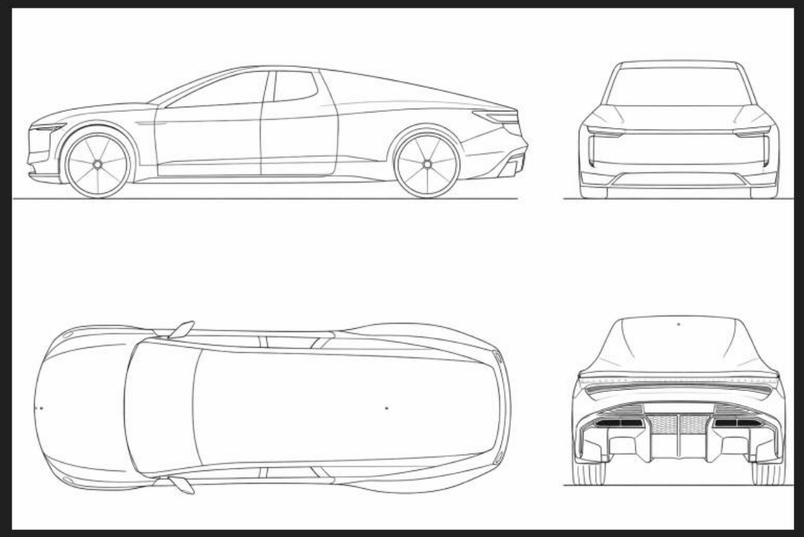


MODEL ITERATING

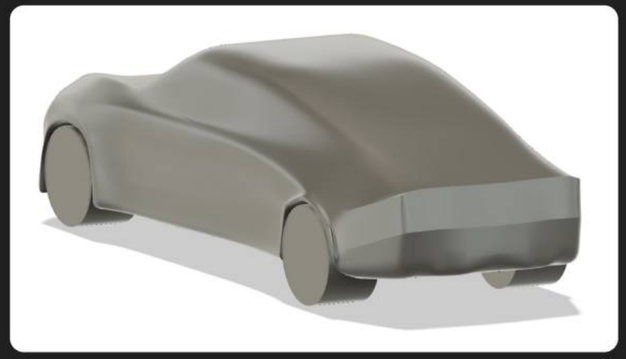


I used carcadblocks.com to find orthographic blueprints of various car models, and adapted/ experimented with them to produce my own.

These were to get a good idea of how to model my car in Fusion, by creating correct canvases to model from within the form tool environment.

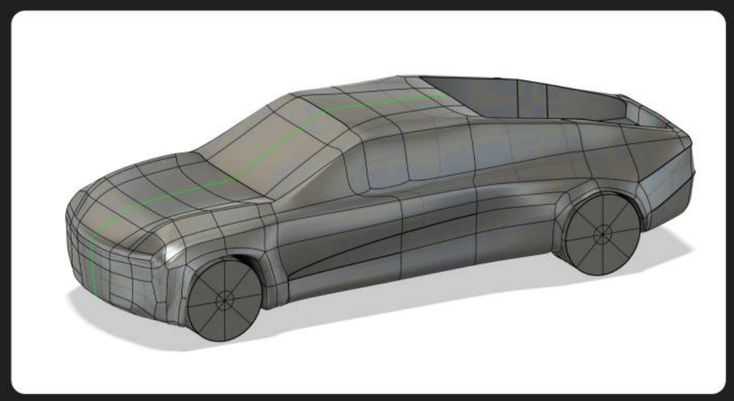


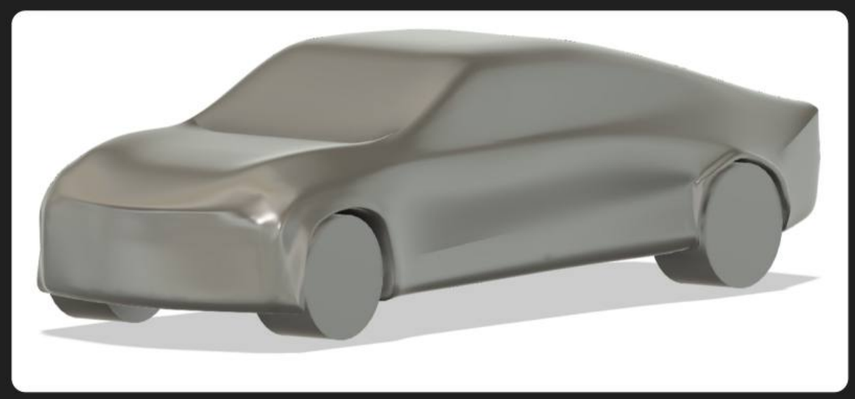




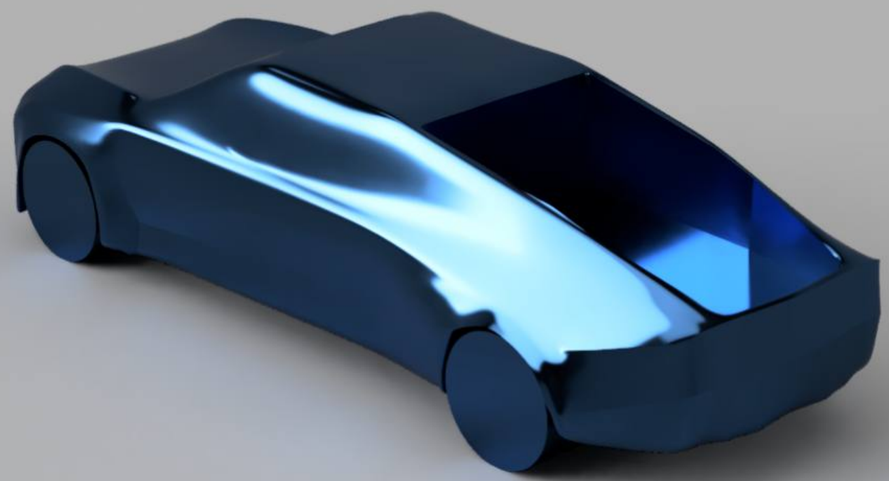


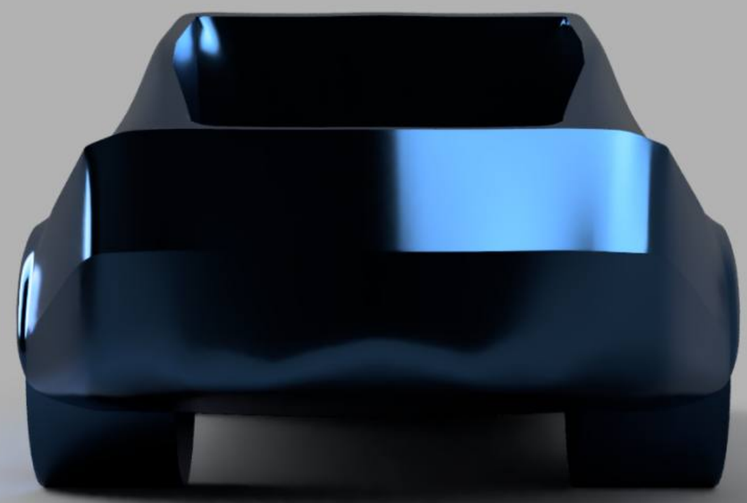
Experimenting with different styles and problems to tackle through CAD
e.g Open vs Closed truck bed



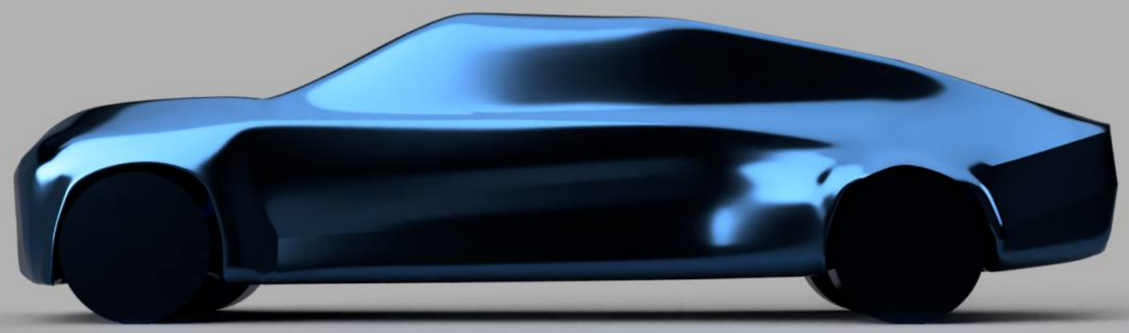


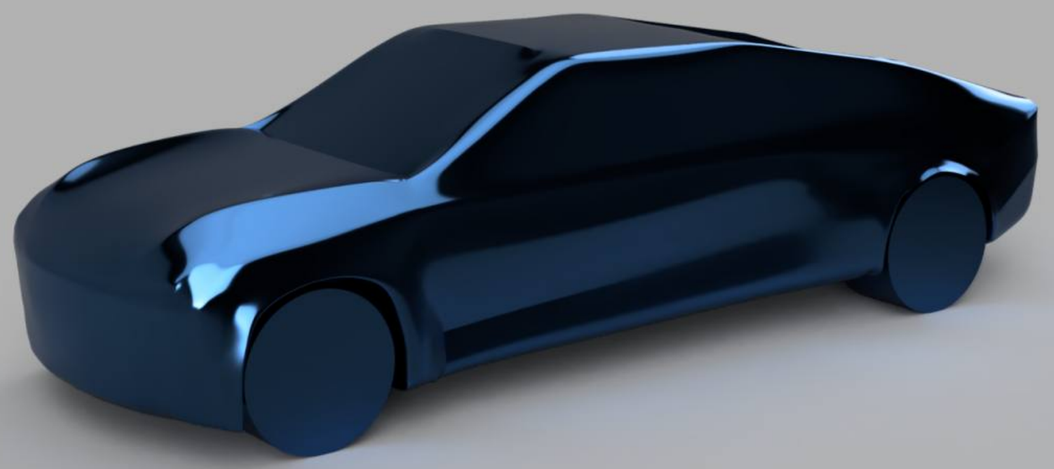




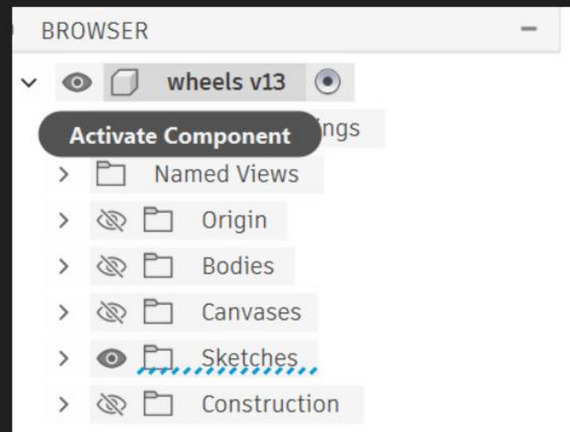


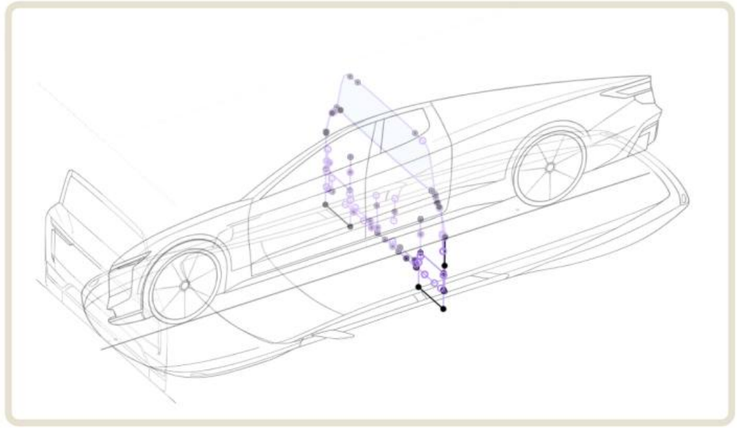
CAD

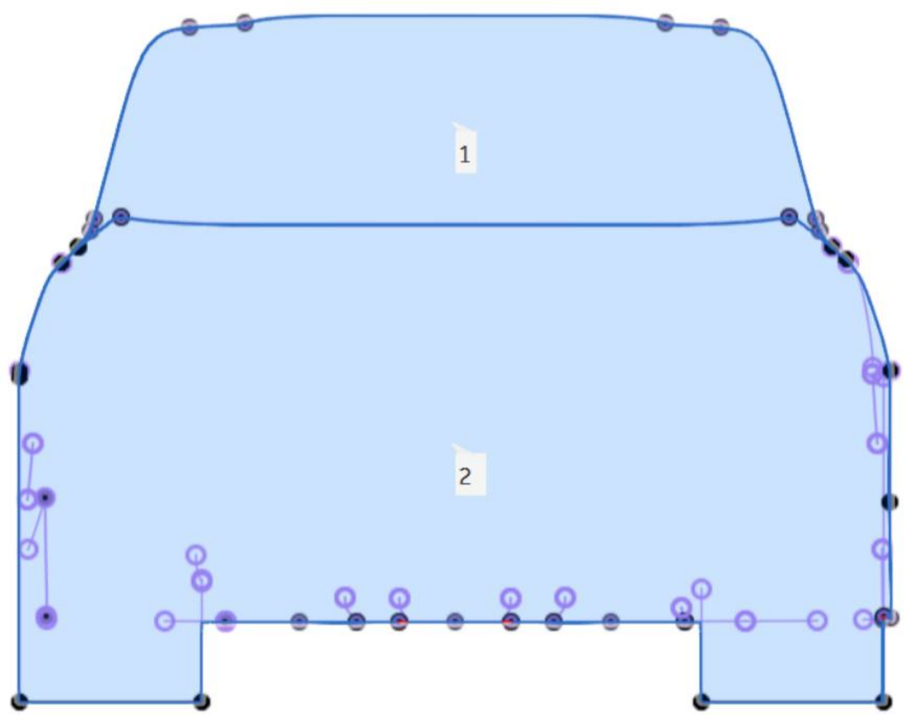


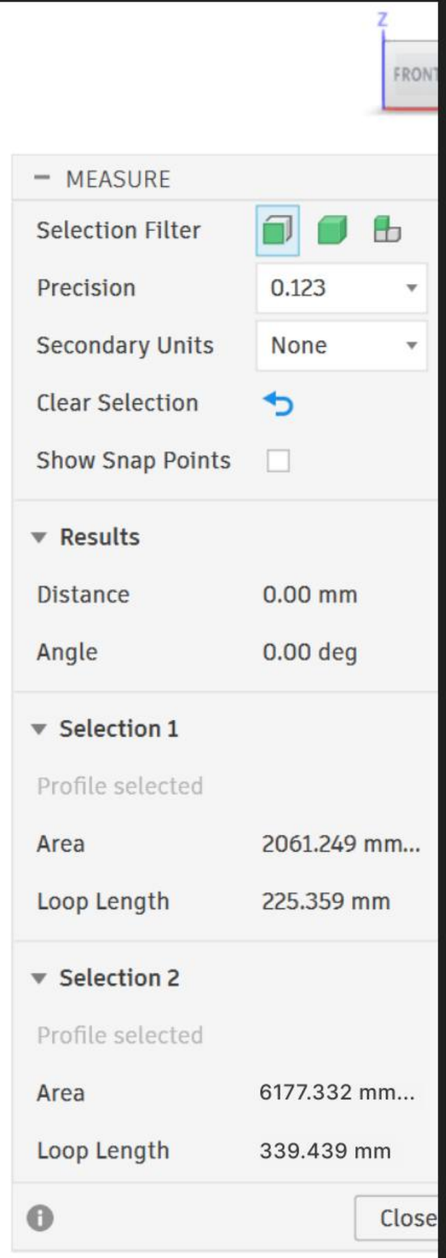


FRONTAL AREA MEASUREMENT





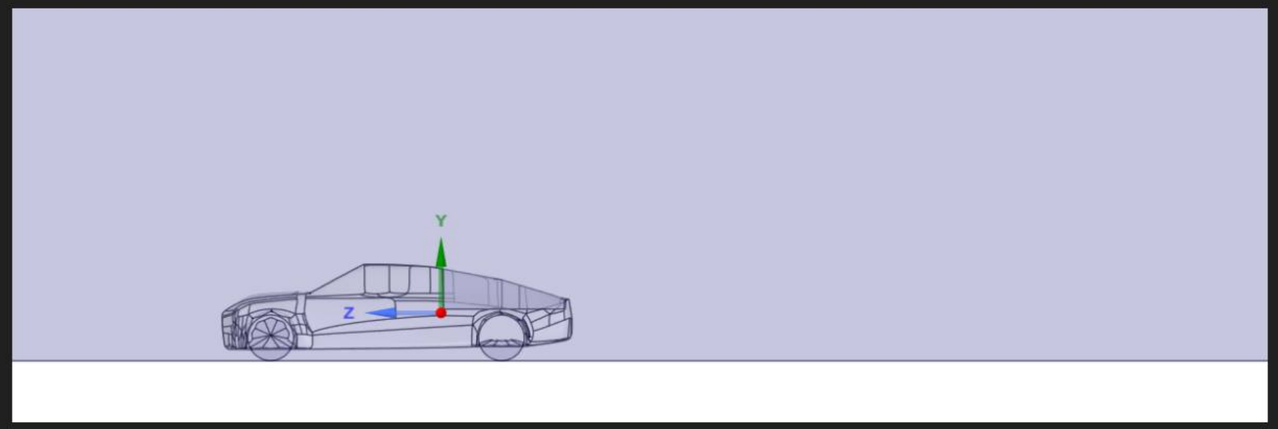


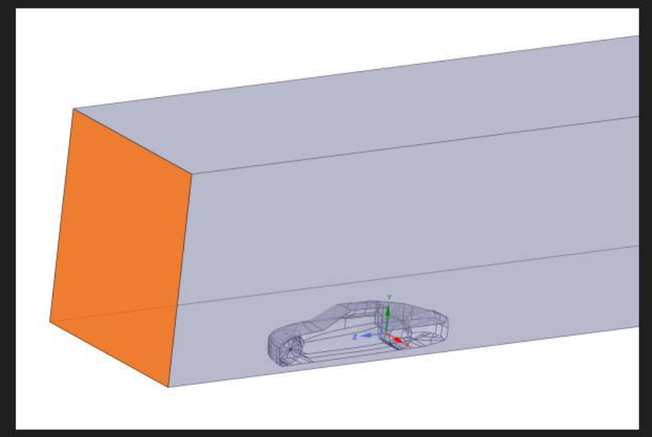


Within Fusion, after creating my model, I projected the car onto a frontal plane to see what faces and edges would come into direct, initial contact with the air. After tidying the projections, I formed two areas - the frontal body which includes the tyres, and the windshield with A-pillars.

The total frontal area came to roughly 4119 mm^2, or 0.004119 m^2

CFD SETUP





```
 513016 cells,     1 cell zone  ...
    513016 polyhedra cells,  zone id
2776682 faces,     7 face zones ...
   2686277 polygonal interior faces,

2136153 nodes,
```

Enclosure dimensions:

- 5x car length behind
- 3x car height above
- 3x car width besides
- 1x car length in front

- Car geometry split
- 1.6 mm gap between car and road surface

2-equation k-epsilon model for turbulence

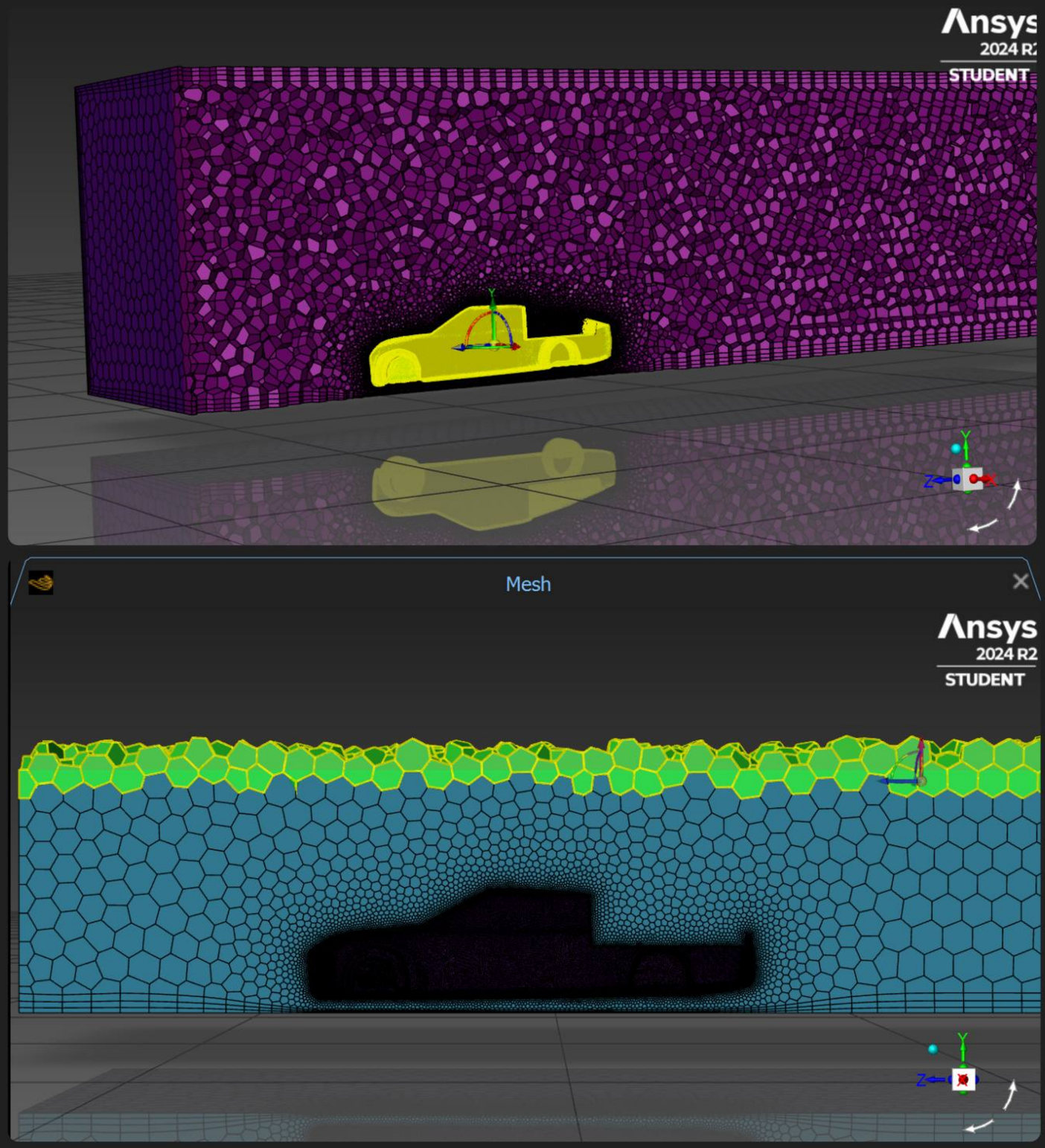
Car wall - No slip shear
Velocity inlet - Normal to boundary, 30.5 m/s (68 mph)
Pressure outlet - Gauge pressure of 0
Road - No slip shear
Symmetry - used to model half the car to speed up process
Walls - Specified shear of 0

Second Order Upwind method used to Turbulent Kinetic Energy and Turbulent Dissipation Rate

This is to ensure that all wakes and airflow affected by the car can be visualised, especially to see airflow attachment to the car body, to see where the streams re-join after leaving the car.

Reference Values	
Area [m^2]	0.004119
Density [kg/m^3]	1.225
Enthalpy [J/kg]	0
Length [m]	0.305
Pressure [Pa]	0
Temperature [K]	288.16
Velocity [m/s]	30.5
Viscosity [kg/(m s)]	1.7894e-05
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300

MESHING



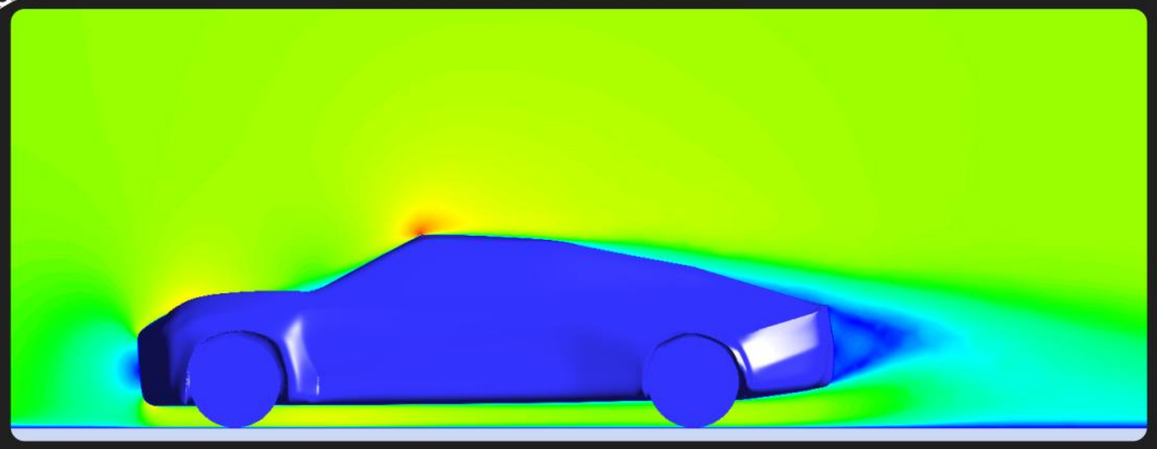
- Target mesh size: 0.001m in local sizing.
- Minimum base size mesh of 0.001m and maximum of 0.02m
- The geometry consists only of fluid regions with no voids
- Fluent solver with polyhedral fill, max cell length of 0.02m
- Clipping planes used to view geometry mesh in XYZ planes

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=\rho g_x-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right)$$

- Computing viscosity and pressure of each cell in the mesh using the Navier-Stokes equation.
- Smaller mesh cells = better accuracy for the model, but comes at a computational cost.
- For the purposes of this modelling, roughly 510,000 mesh cells should suffice, as it is a low-fi CAD model. For better accuracy in future, use a refined CAD model and compute with finer meshes.

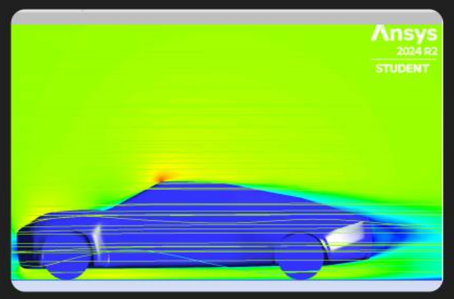
2D PRESSURE & VELOCITY

VELOCITY PLOT

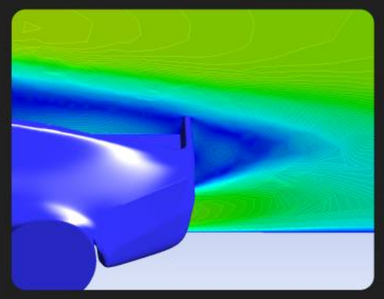


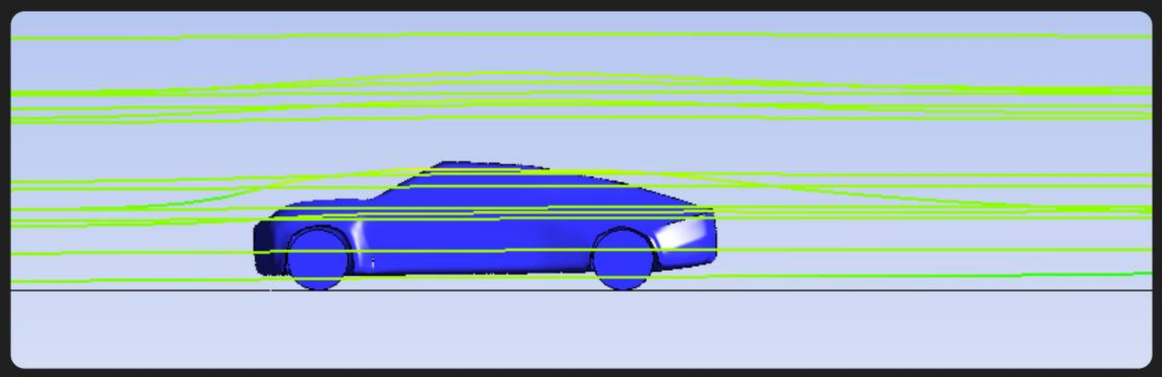
The velocity plot shown shows that the highest region of velocity is above and below the car, indicated by the green-yellow regions.

The lowest are blue, at the front and rear of the car. At the rear is the wake, where the boundary layer air separates from the car surface, and is drastically slowed down. High-velocity air is what mixes and pulls the slow/stagnant air/ This is the chaotic airflow that needs to be minimised.

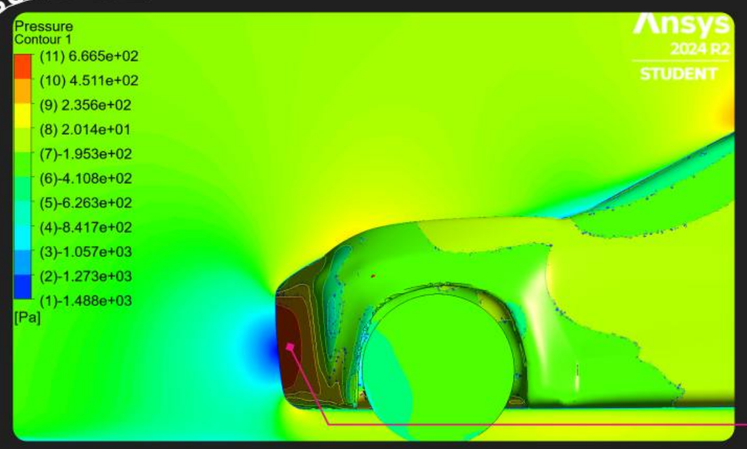


Streamlines show the air is curving up across the front wheels and then down low by the rear. This longer path suggests it is colliding with the body more sharply and being redirected.



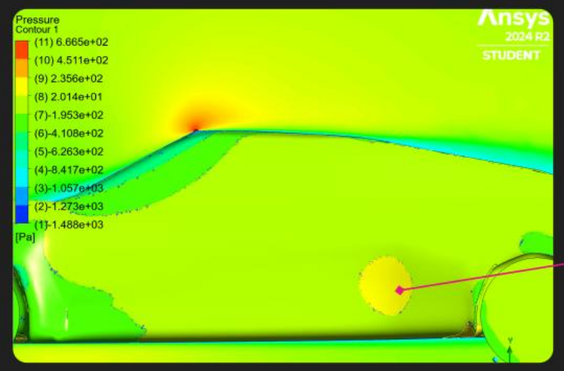


PRESSURE PLOT

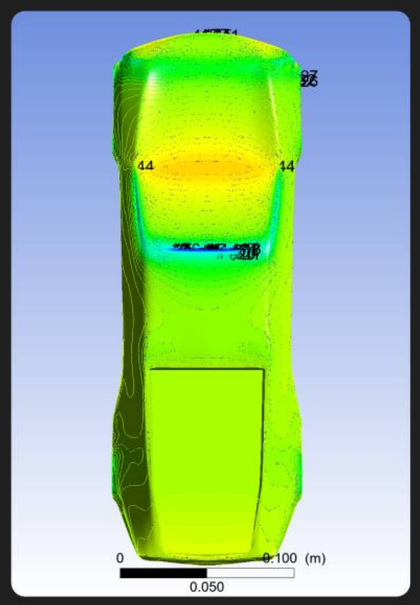


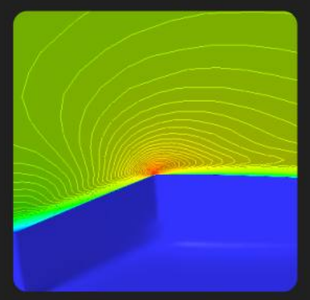
Fast air has low pressure, and slow air has a high pressure. This is in principle, how Bernoulli's equation is used to design wings, diffusers and most aero components in vehicles.

There is high pressure in the frontal region of the car, as the slight angle forces air around.



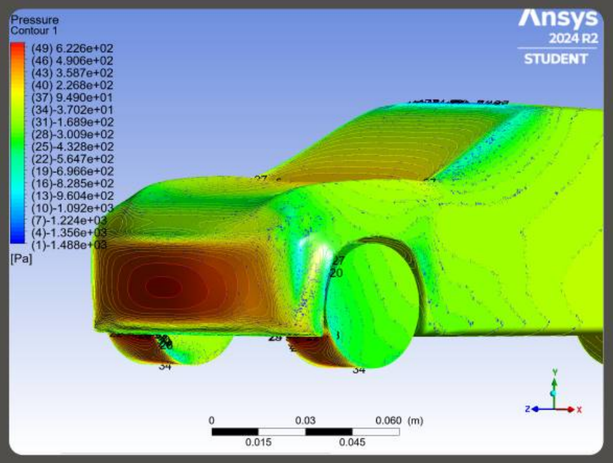
There is high pressure at the junction between the windscreen and roof of the car. This is likely due to the fact that I creased the edge, as most pick-up trucks have a distinct transition between the two.
There is also a region of high pressure in the body panel towards the rear, because of the contoured shape that was redirects airflow down.





The roof flow accelerates strongly near the front (yellow region) and forms long shear or vortex streaks along both roof shoulders (blue bands), with vectors showing some outward drift toward the sides, which points to 3D vortices that can feed a larger rear wake; smoothing the A pillar to roof transition and increasing roof edge radii should weaken these streaks and help rear pressure recovery.

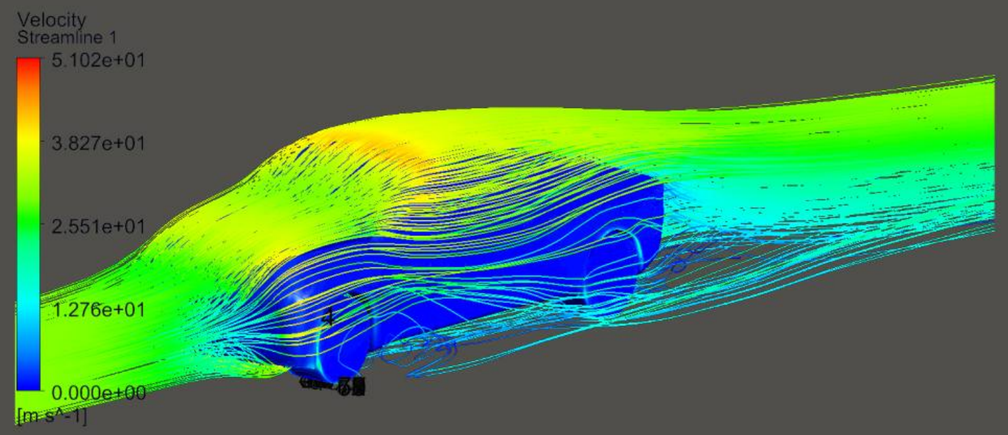
3D PRESSURE & VELOCITY

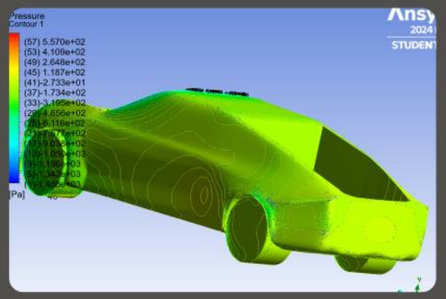


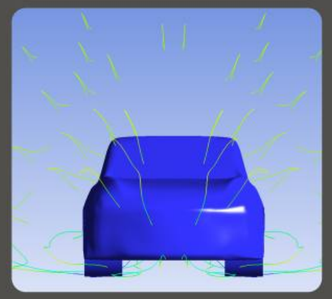
The front of the car is what comes into contact with the most air and the most direct, creating a stagnation point indicated by the concentric red circle at the front. The convex shape does force some air above and below, but the pressure is largely here.

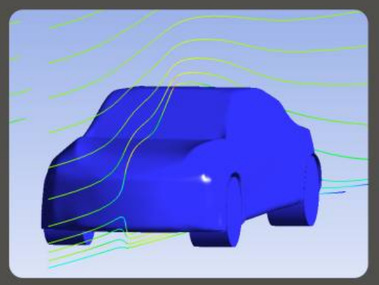
The velocity streamline indicates turbulent airflow just after the wheel arches, drastically slowing the air down and creating messy wakes behind the car. This will hurt the efficiency and may need to lead to extra road noise when driving at high speeds.

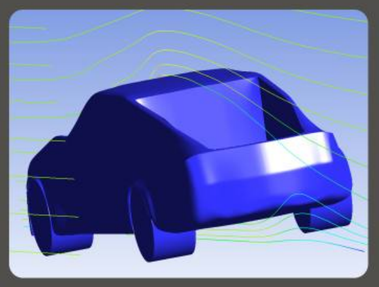
Majority of air hitting the front is redirected around the car, through body contours or over the top.



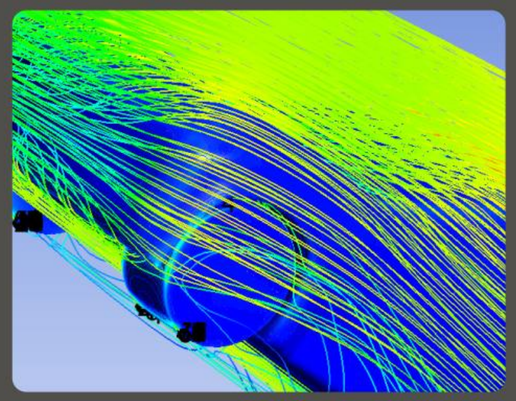








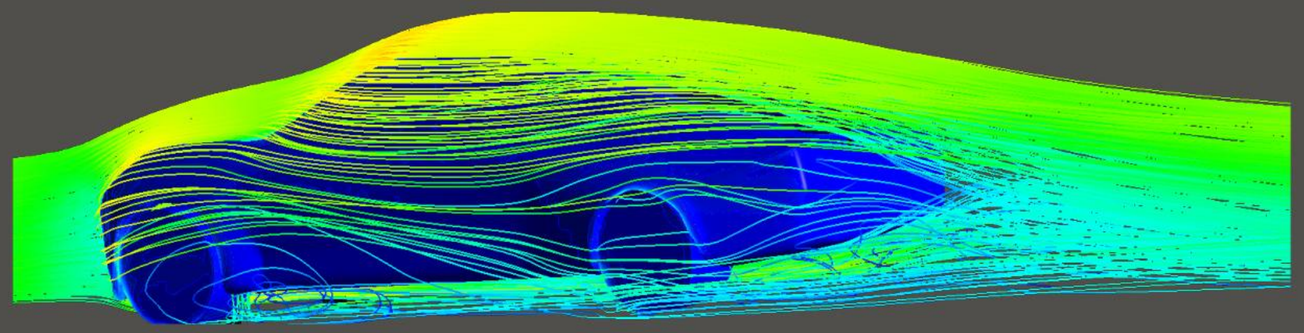
Front, side and rear views of the velocity and pressure. High pressure in the body contour.
The bed of the truck does act as an air cushion, as the majority of the air flows right over it



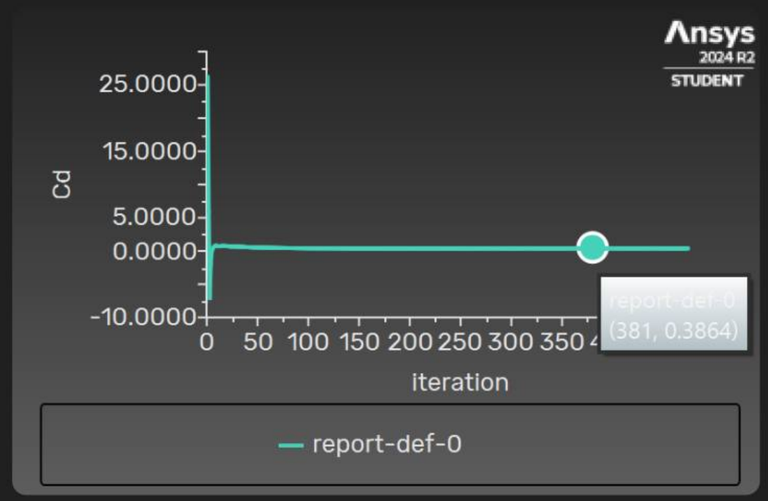
Turbulent Eddy Dissipation describes to us where the most energy conversion to thermal energy is occurring. As indicated, the majority of this is at the creases of the wheel arches, likely due to them being fairly sharp, and protruding from the body.



Although the air does seem to join back together somewhat cleanly at the rear of the car, a lot of it is experiencing turbulence beneath the wheel arches, slowing it down. Whilst some is successfully rerouted to the rear, lower down the air is stagnating beneath the car, contributing to a lower lift coefficient, hence decreasing downforce perhaps unexpectedly.



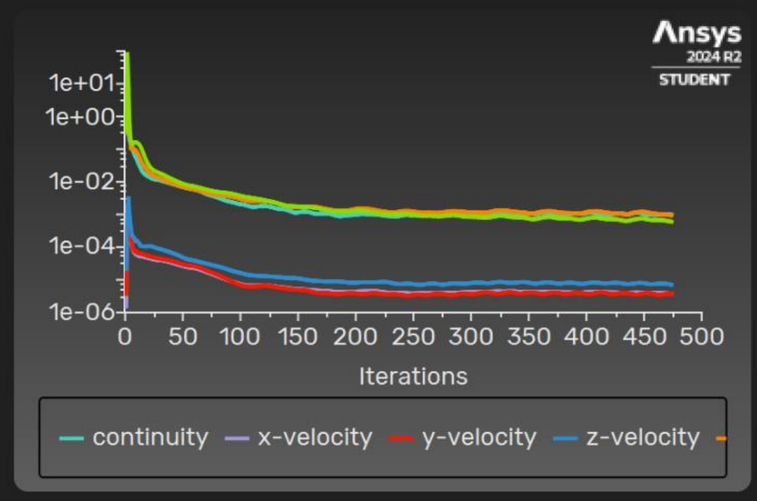
AERODYNAMIC FORCES



Drag coefficient (Cd) : 0.3864



Lift coefficient (CI) : 0.2396



Miscellaneous parameters

$$
\begin{aligned}
F_{\text{drag}} &= \frac{1}{2}\rho v^2 A C_d \\
&= \frac{1}{2}(1.225)(30.5^2)(0.004119)(0.3864) \\
&= 0.907 \text{ N}
\end{aligned}
$$

$$
\begin{aligned}
F_{\text{down}} &= -\frac{1}{2}\rho v^2 A C_l \\
&= -\frac{1}{2}(1.225)(30.5^2)(0.004119)(0.2396) \\
&= -0.562 \text{ N}
\end{aligned}
$$

These calculations are based off of the frontal area value for my 1:17 scaled model. If we extrapolate:
0.004119 * (18)² = 1.3578 m² frontal area. New forces:
Drag: 293.87N
Downforce: -182.09N

DBX 707

[14]



Drag coefficient comparable to these cars

CYBERTRUCK

[8]



500 iterations of the model were ran to produce these results.

Continuity represents how well the simulation is suited to repeated runs/ conversions.

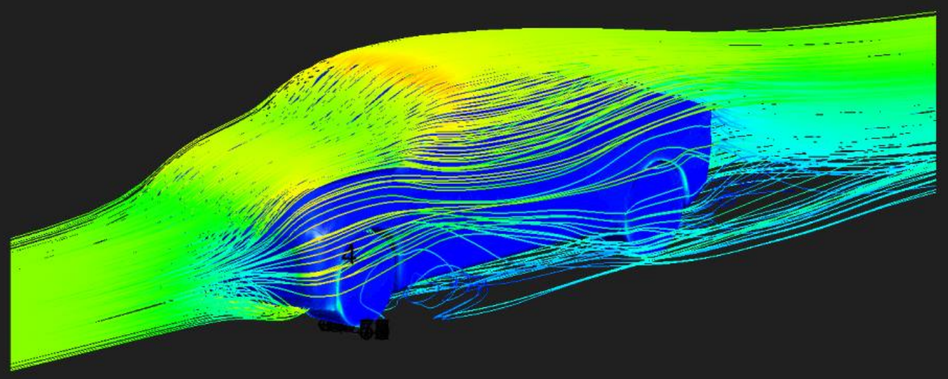
XYZ velocity are the components in each direction.

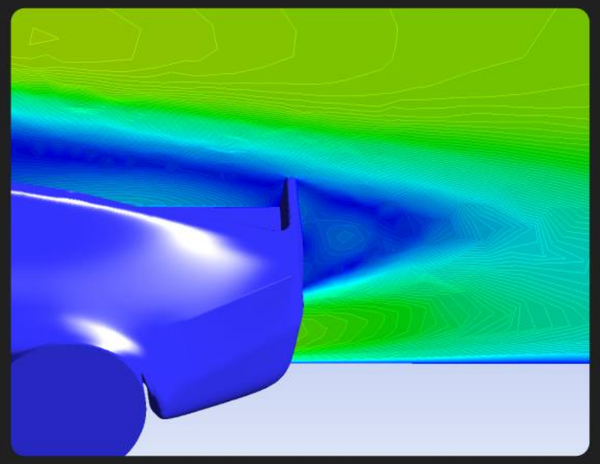
Orange is the Turbulent Kinetic Energy (k)

Green is the Turbulent Dissipation Rate (Ɛ)

ANALYSIS OF RESULTS

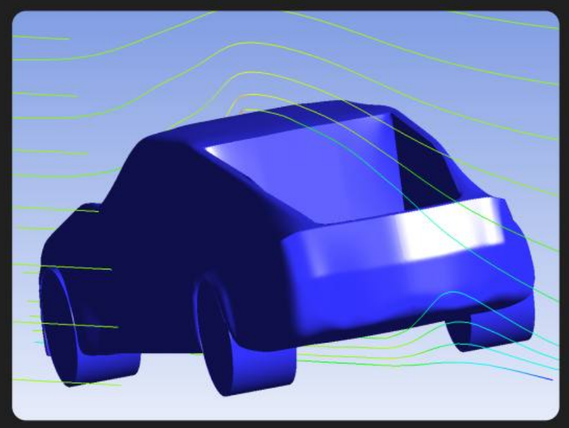
SUCCESSES



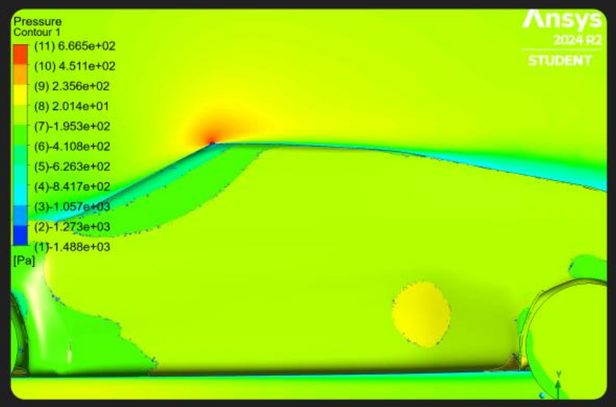


Air flow redirected well for the most part, around the body into a tapered wake at the rear. Relatively low drag coefficient of 0.38

Sloped bed acts as an air cushion and allows air to flow over it. Slight diffuser does channel the air out with the rest of the boundary airflow.



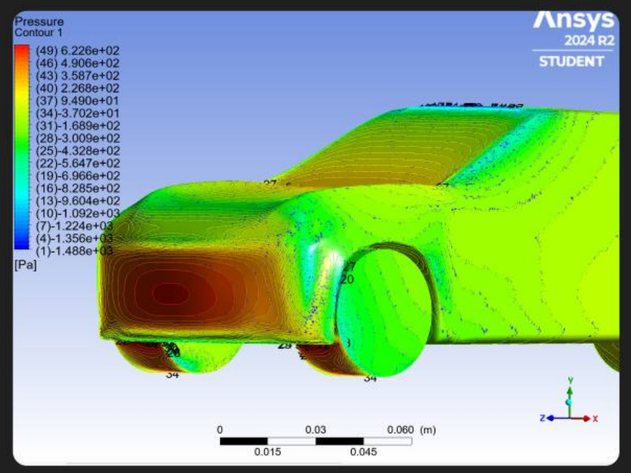
DOWNFALLS

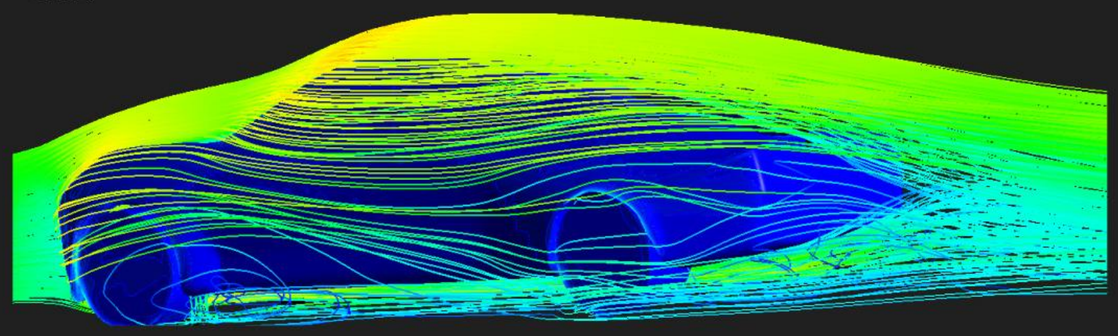


Sharp windscreen/ roof edge has formed a high pressure region, resulting in some drag. High velocity as well.

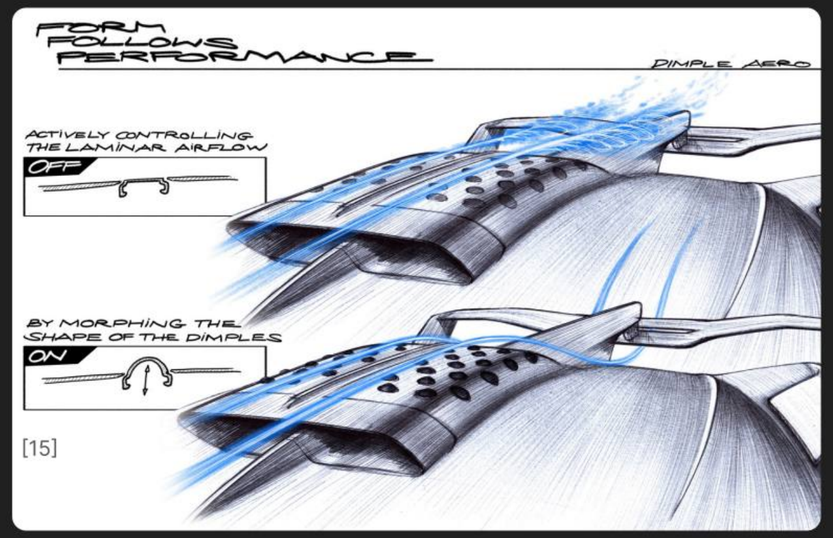
Frontal area has high pressure, but not a lot of downforce created (CI = 0.24).

Sharp wheel arches are a point of kinetic → thermal energy conversion, as well as a source of turbulent air.





FUTURE PROPOSALS AND IMPROVEMENTS

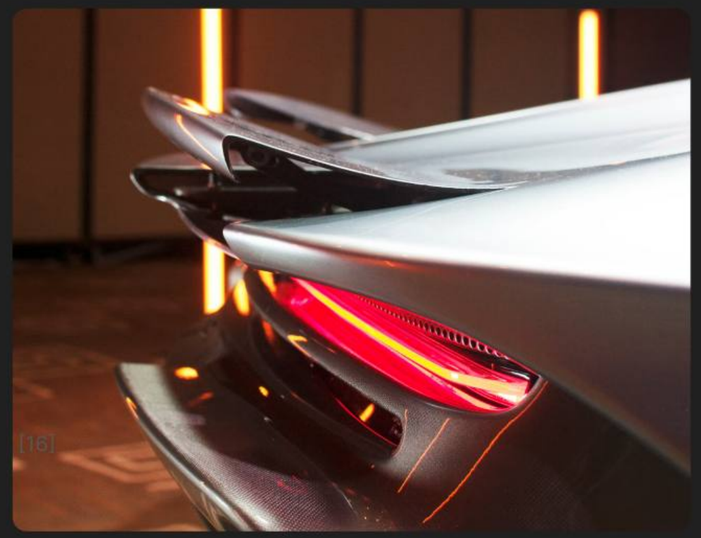


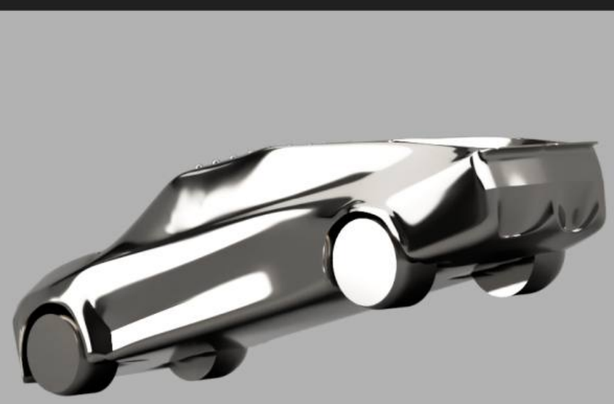
[15]

The Bugatti Bolide utilises a golf-ball dimple shaped structure on the roof of its air intake. This is designed to smoothen out turbulent airflow that appears at high speed, quoted to work best above 74 mph. Whilst pick-up trucks do not always reach those speeds, they are however, driven long distances at high average speeds, where this system can still prove beneficial.
Much like a golf ball moves smoothly through the air, these dimples are active; deployable when needed.

The McLaren Speedtail features an active aero spoiler, where the body panels morph to create the image shown. This could be integrated within the tailgate of my pick-up truck, but deployable as a separate piece rather than a morphing panel to reduce cost.

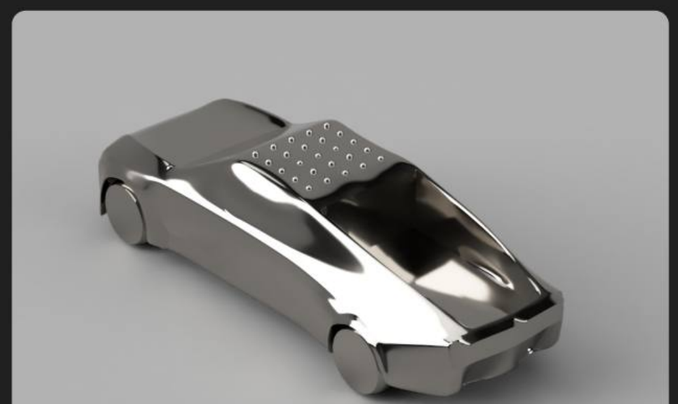
Active parts have the benefit of you choosing not to deploy them, which is crucial if you are carrying bulky or dangerous items that may get in the way of such components.


[16]



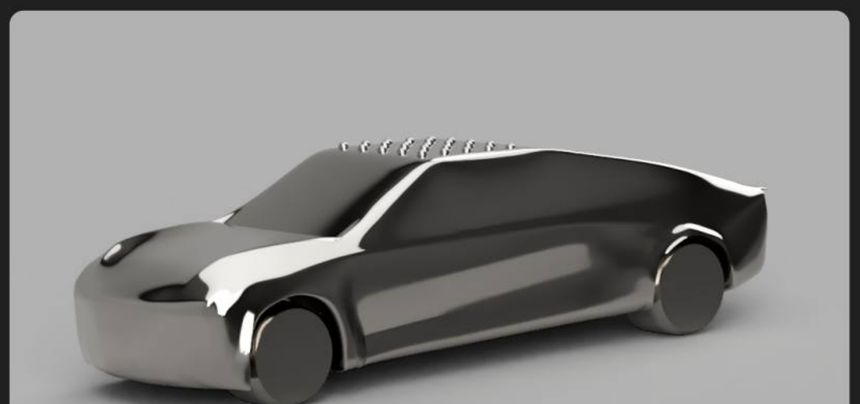
Enhanced rear diffuser to better-combat the turbulent undercarriage airflow, caused by the wheel arches and body contours. This should decrease drag and simultaneously increase downforce. Whilst downforce is not imperative for a truck carrying heavy loads, one cannot assume that a pick-up truck will always be carrying heavy loads.

Combined with air channel duct from vents in the wheel arches to accelerate air past the car, inspired by the Lotus Evija.



Golf-ball dimples added to the roof for improved aero at high speeds, courtesy of the Bugatti Bolide.

A rear wing has also been added, with a notch in line with that of the accentuated rear diffuser. This should increase the time it takes for the boundary layer air to meet again, creating a larger “teardrop” wake profile.



Windscreen/roof meeting point has been rounded out and uncreased. The angle has been adjust to preserve the aesthetics of pick-up trucks, whilst improving airflow over the top of it.

The wheel arches has been tapered more, with less sharp edges and protrusions. They also transition into the body contours more gradually, to avoid those high pressure spots seen during the CFD analysis.

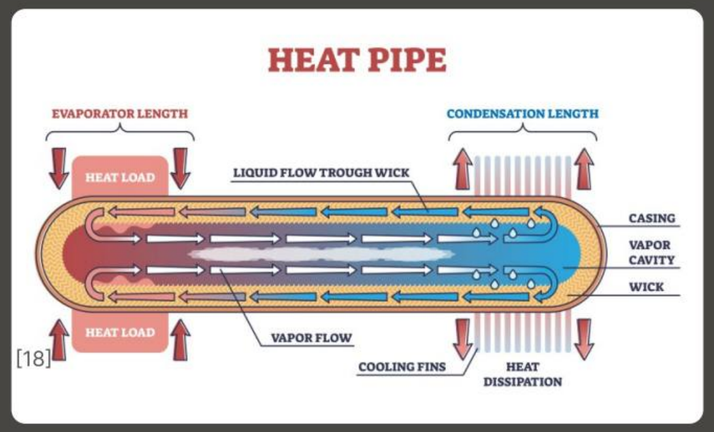
PIPELINE RESEARCH & ITERATING

Objective: Develop a liquid-cooled plate design that balances temperature uniformity, maximum cell temperature, and pressure drop, based on benchmarking of existing EV and electronics cooling solutions.

EV's require batteries and motors instead of engines or fuel tanks. Although they do not reach the same temperature, batteries do need to be cooled down to ensure they are within their optimal operating temperature range, 20-40°C.

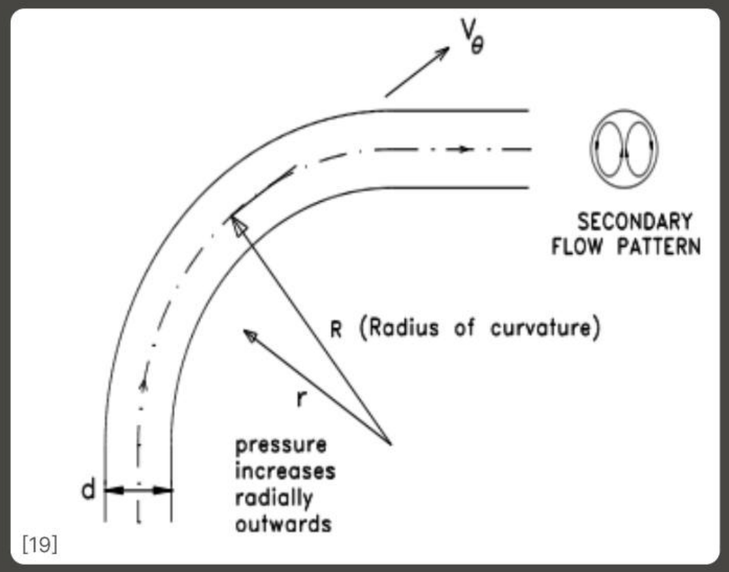
Pipe cooling systems, in their base form, rely on channels made of thermally conductive material, surrounding the target object. There also needs to be a system to remove the heat from the system.

Often, this is in the form of liquid cooled pipes, whereby a pump is connected to an inlet to pump coolant through, and an outlet to retrieve the liquid, cool it down, and cycle it through.



[18]

Computers used heatsinks to distribute heat - small fins are packed in a dense structure, as this increases surface area, thereby increasing the rate at which thermal energy can be removed. Graphics card utilise a heat pipe system instead - passive thermal conductor process. A small amount of liquid is contained within these pipes. Heat causes the liquid to vaporise, transporting the the now-vapour away from the heat source. At the other end, it cools down, thereby condensing, and returns to the heat source.



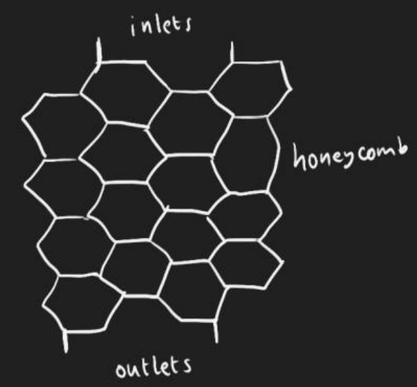
[19]

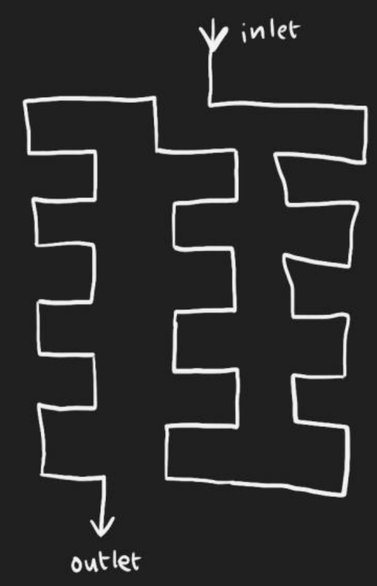
A cross-stream pressure gradient has to exist in order to provide the centripetal force for circular motion with the pressure is higher on the outside of the bend [19]. A larger radius for our pipes is favourable, as it will be less restrictive of the liquid's flow. Too many bends/sharp bends will drastically slow down the flow, and narrower pipes' increased flow velocity leads to greater frictional losses, and therefore pressure drop.

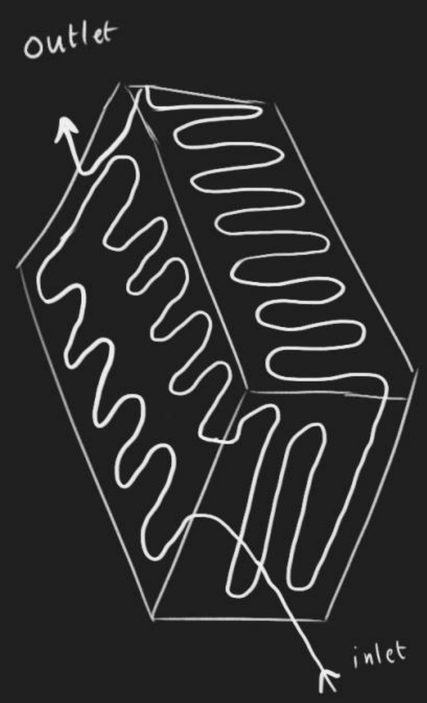
The same applied to lengthening pipes- pressure loss, and therefore you will need a more powerful pump to compensate, or have multiple inlets.

A balance between the two is required to achieve optimal flow rate, which will be explored through a series of design iterations.









DESIGN RATIONALE & BENCHMARKING

Fractal/Biomimetic Branching

In nature, species adapt or die out. The world is old, and so what species still remain are living examples of optimisation.

For example, the veins in leaves to distribute nutrients. The complex structure allows for cells to individually receive/send nutrients to the rest of the tree, because each branch of veins has another set of veins. They maintain controlled hydraulic pressure.

Similarly, the human body's circulatory system supplies oxygen uniformly. Width of channels (vessels) depends on how central it is. Veins are typically narrower while arteries are larger in diameter.



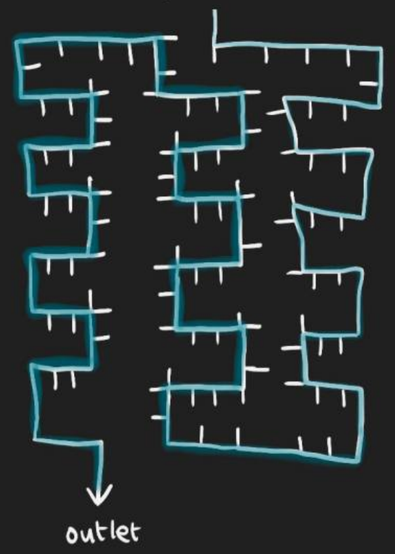


Shorter branch lengths reduce longitudinal heating. Improved peak temperature control due to the complex and denser network of pipes/veins in the system.

A good example of this in a heat distribution system are elephant ears. They flap their ears for communication, but also for thermal regulation. The large surface area of their ears allows them to radiate heat when they are too warm.

Iterated my designs for greater surface control. The first one makes uses of passive fins for additional thermal withdrawal to the active channels.
The second one is inspired by branching in nature, where all the channels flow smoothly with few sharp turns.

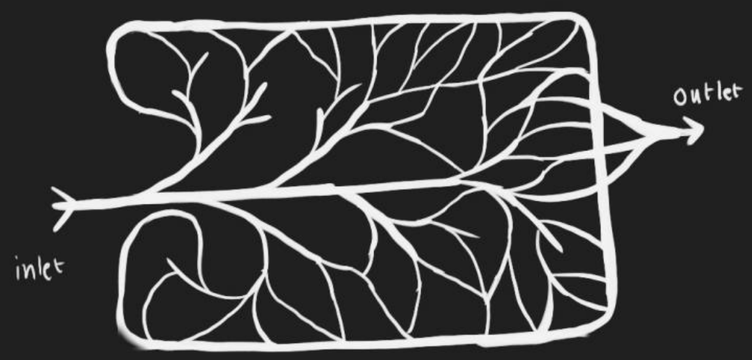
Serpentine



Pro's: High surface area/contact area with battery pack. Uniform channel sections make hydraulic resistance easier to account and compensate for.
Constant channel width means we can easily machine this with a CNC mill, and bond over the cooling plate. Flat shape will sit comfortably on battery packs.

Cons: Numerous 90° turns leads to velocity loss and minor pressure loss.
One long pipe is not great because longer pipes will require more force to keep the coolant flowing at an optimal velocity.

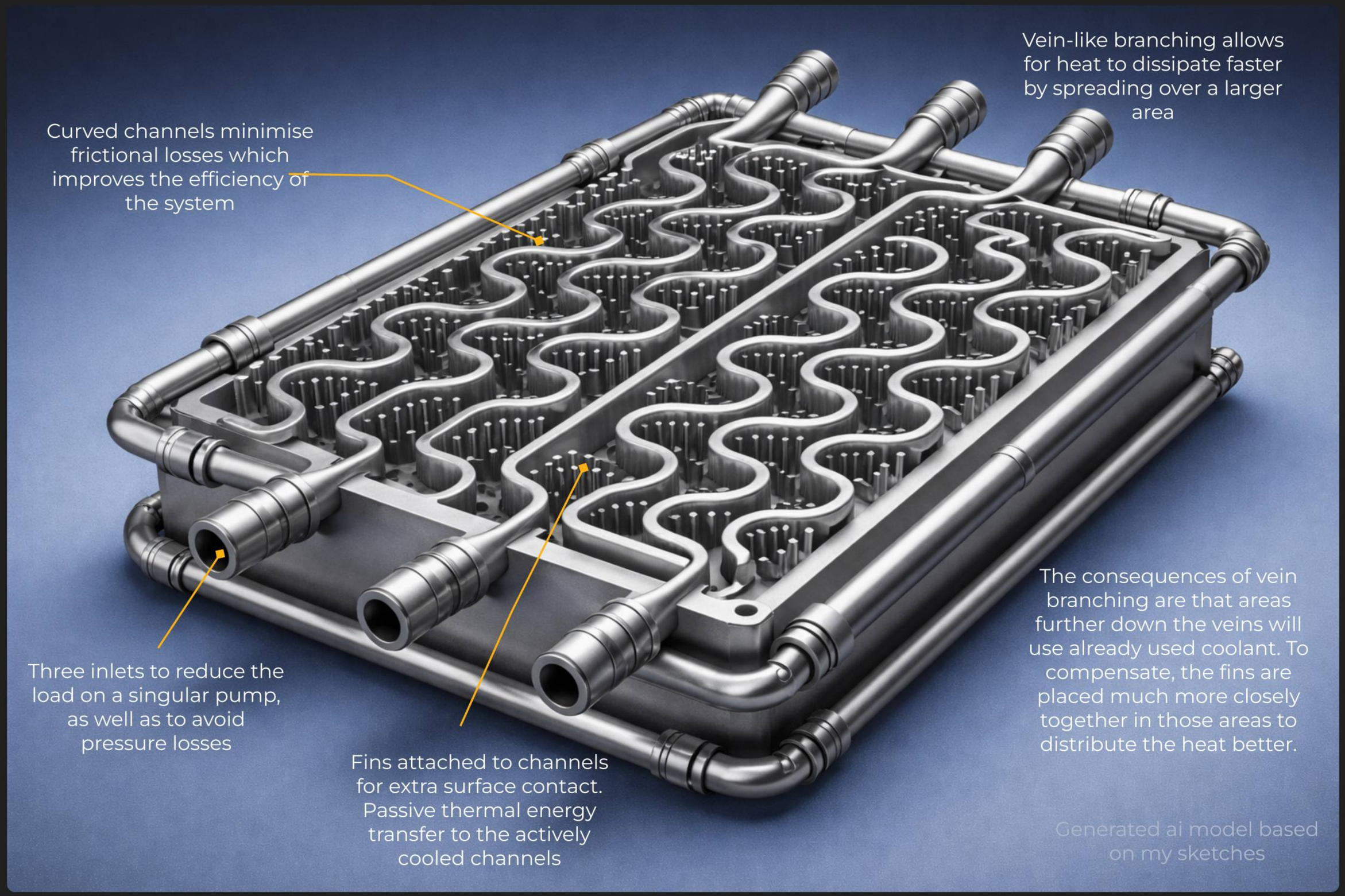
Leaf veins



Pro's: Great temperature uniformity. Many shorter paths for coolant to flow through - active cooling covers great surface area.
Branches are smooth. Few sharp turns and bends in the pipe so frictional losses are minimal.

Cons: Some regions may get less flow if modelled improperly, as the coolant will try to flow through the shortest path.
As coolant may get reused for other paths before reaching the outlet, there may be hotspots in the plate.
Complex to manufacture due to numerous small paths and curved nature. Mixing losses at junctions.

BATTERY CONCLUSION



CONCLUSION

Velocity

- Low velocity wake behind the car created.
- High velocity at the windscreen-roof meeting point.
- Turbulent air by the wheel arches caused by sharp and creased edges.
- Low velocity at the front.

Airflow

- Sloped bed acts as an air cushion and allows air to flow over it. Slight diffuser does channel the air out with the rest of the boundary airflow.
- Low velocity airflow beneath the wheel arches.
- Most air is guided by contours and side panels to exit at the rear of the car.

Next steps:

- Wind tunnel test to compare results with CFD calculations.
- Model pipeline accurately.
- Calculate heat transfer of pipe cooling system.
- Calculate pressure and velocity losses in the pipe system.

Pressure

- High frontal pressure. Flat is not convex enough; it needs to be more rounded to dissipate the air around it.
- Medium pressure in the body contour, as it merges with the wheel arches too sharply.
- Low pressure below the stagnant point in front creating downforce.

Forces

- Drag Coefficient: 0.3864
- Drag Force: 293.87 N
- Lift Coefficient: 0.2396
- Downforce: -182.09 N

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